

The Effect of Boarding Strategies on Aircraft Boarding Efficiency

A Simulation-Based Study with an Exact Longest-Path (Max-Plus) Theory | Author: Zihao Huang | October 2026

Abstract

This study compares three aircraft boarding strategies — *random*, *back-to-front*, and *zoning* (5 zones) — using a single-aisle simulation. Contrary to intuition, **random boarding is the fastest and back-to-front is the slowest**, and the gap widens with stow time and cabin size. We derive an **exact theoretical model** that computes the boarding time directly (without tick-by-tick simulation) as the length of a **longest path** in a max-plus recurrence; it reproduces the simulation **bit-for-bit** (verified over >25,000 cases), and yields provable lower bounds. The theory — corroborated by measured mechanism metrics (parallelism and blocking) — explains *why* clustered boarding is slow: consecutive same-row passengers stack stowing weights on a single aisle cell and lengthen the critical path.

1. Background and Research Question

Boarding time affects on-time performance and operating cost. Airlines often use "zone boarding" or "back-to-front boarding", but whether these beat random boarding is debated. We test:

Hypothesis H: Organized strategies (back-to-front / zoning) board faster than random boarding.

2. Methods

2.1 Simulation model

- Single-aisle cabin; passengers enter from the front and walk to their row.
- On reaching the row, a passenger stows luggage, occupying the aisle for **S** ticks; followers cannot pass.
- One cell of forward motion per tick; no overtaking; one passenger enters per tick.
- Total ticks until all are seated = boarding time (smaller = faster).

2.2 Variables

Type	Variable
Independent	Boarding strategy (Random / Back-to-front / Zones-5)
Dependent	Total boarding time T (ticks)
Control (baseline)	Random boarding (normalized to 1.00)
Controlled	Rows R ; seats per row (6); entry rate (1/tick); repeats (20)

2.3 Mechanism metrics

The simulator records, each tick, the number of passengers **stowing simultaneously** (parallelism) and the number of **blocking events** (walkers stuck behind a stower).

2.4 Theoretical model: an exact longest-path (max-plus) formulation

We model boarding as a **discrete-time single-lane exclusion process** and derive an **exact recurrence** for T . Let

- cells be $1..R$; passenger i ($i = 1..N$) enters in order; r_i = target row; S = stow time;
- P_i^c = the tick at which passenger i reaches cell c ;
- $j = j_i(c)$ = the last passenger before i that reaches cell c .

Passenger i reaches cell c at the **later** of two times: its own pace, and the moment cell c becomes free:

$$P_i^c = \max(P_i^{c-1} + 1, B_i^c), \text{ where } B_i^c = \{ P_j^c + 1 + S \text{ if } r_j = c \text{ (j stows at row } c); P_j^{c+1} \text{ if } r_j > c \text{ (j passes through)} \}$$

with the exact entrance conventions (entering is its own tick; right after entering one may move an extra cell):

$$P_i^1 = \max(i, B_i^1), \quad P_i^2 = \max(P_i^1, B_i^2), \quad P_i^c = \max(P_i^{c-1} + 1, B_i^c) \text{ for } c \geq 3.$$

The finish time of passenger i and the total time are

$$F_i = P_i^{r_i} + [r_i \geq 2] + S - 1, \quad T = \max_i F_i.$$

Interpretation. Every P_i^c is a *max of two terms* ("my own pace" vs "waiting for the person ahead"), so T is the length of the **longest chain of waits** — a **longest path** (last-passage percolation) value on the space–time lattice.

2.5 Lower bounds

Proposition 1 (entrance bound). $T \geq N$. Proof: at most one passenger enters per tick, and all N must have entered.

Proposition 2 (occupancy bound). $T \geq (\sum_i r_i + N \cdot S) / R$. Proof: passenger i occupies r_i cell-ticks while walking and S while stowing, so total occupancy is $\sum_i r_i + N \cdot S$; at most R cells can be occupied in any tick. For 6 seats per row ($\sum_i r_i = 3R(R+1)$): $T \geq 3(R+1) + 6S$.

2.6 Data collection

Strategies with randomness (Random, Zones) are run **20 times and averaged**; Back-to-front is deterministic. Results are reported relative to random (baseline = 1.00).

3. Results

3.1 Main result ($R = 30$ rows, $S = 3$)

Strategy	Average boarding time (ticks)	Relative to random
Random	331.9	1.00 (baseline)
Zones (5)	402.6	1.21
Back-to-front	627	1.89

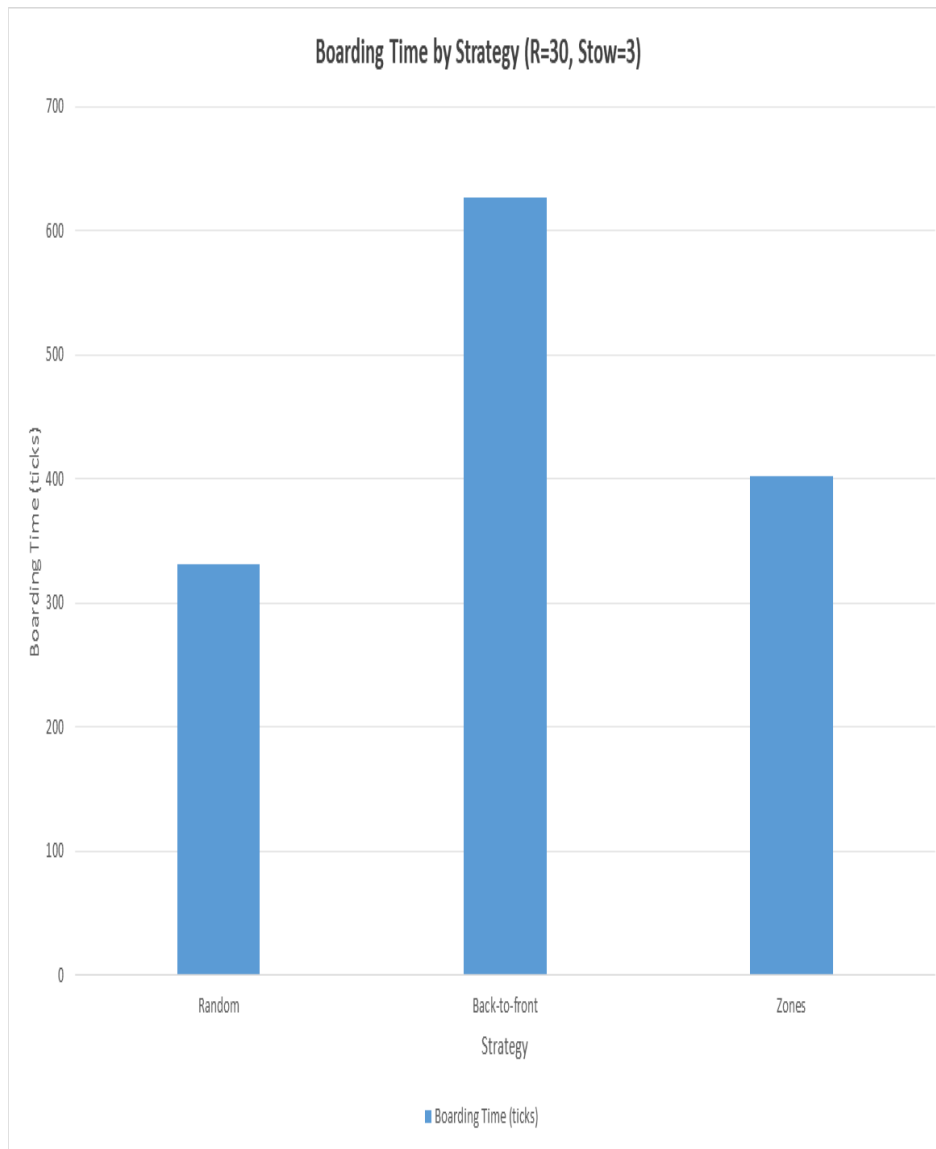


Figure 1. Boarding time by strategy (R=30, S=3): random fastest, back-to-front slowest.

3.2 Sensitivity analysis

(a) Stow time (fixed R = 30).

S	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
1	238.9	325	249.9	1.36	1.05
3	331.9	627	402.6	1.89	1.21
6	471.5	1080	633.8	2.29	1.34

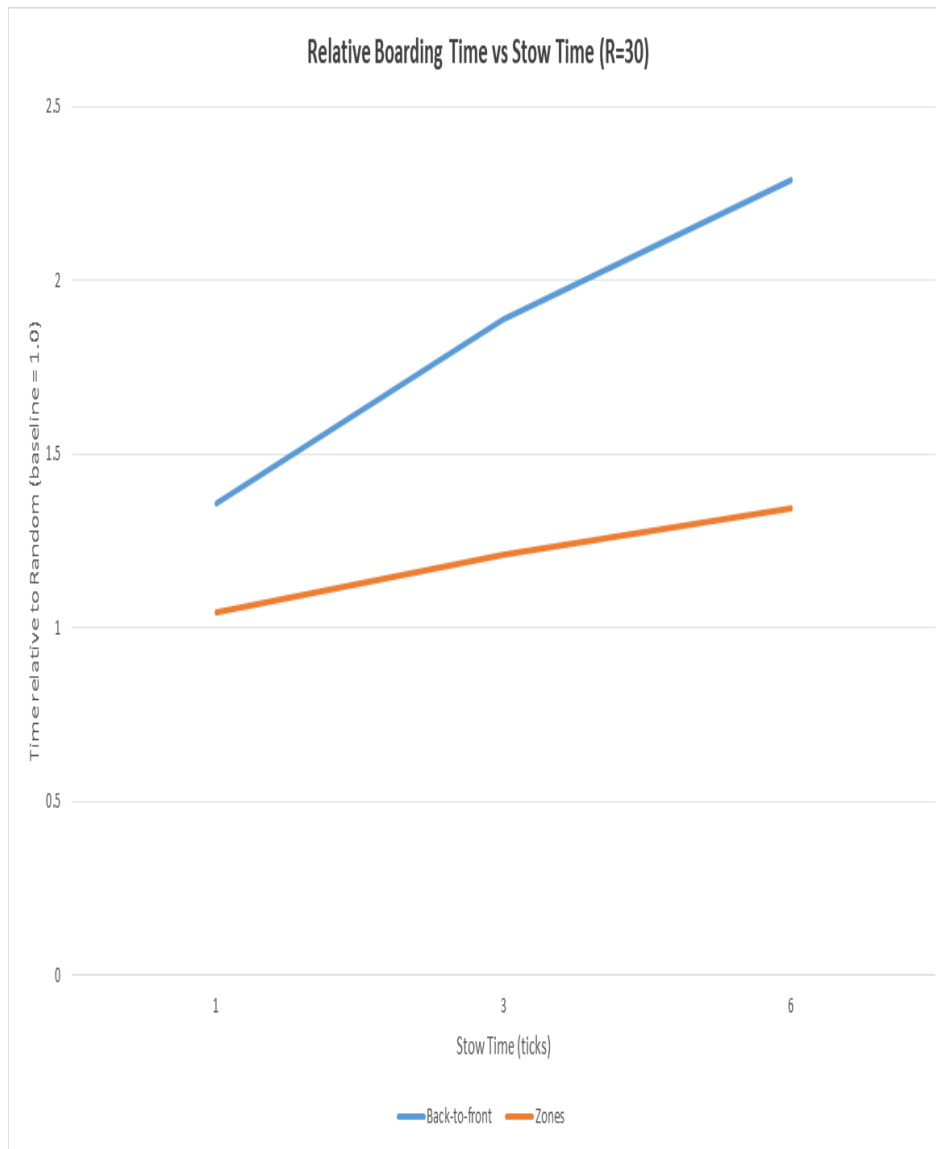


Figure 2. Relative boarding time vs stow time (R=30).

(b) Number of rows (fixed S = 3).

Rows R	Random	Back-to-front	Zones (5)	Rel. random (back)	Rel. random (zones)
20	234.3	417	288.7	1.78	1.23
30	331.9	627	402.6	1.89	1.21
40	423.6	837	507.3	1.98	1.20

3.3 Why: the empirical mechanism and its exact theory

Empirical mechanism. The measured mechanism agrees with the theory: the fastest strategy is also the most parallel. The average number of passengers stowing at the same moment is **1.09** for random versus **0.57** for back-to-front, and the total number of blocking events is **~950** versus **6,525** (Figure 3) — random lets many rows stow simultaneously, while back-to-front forces one-by-one stowing.

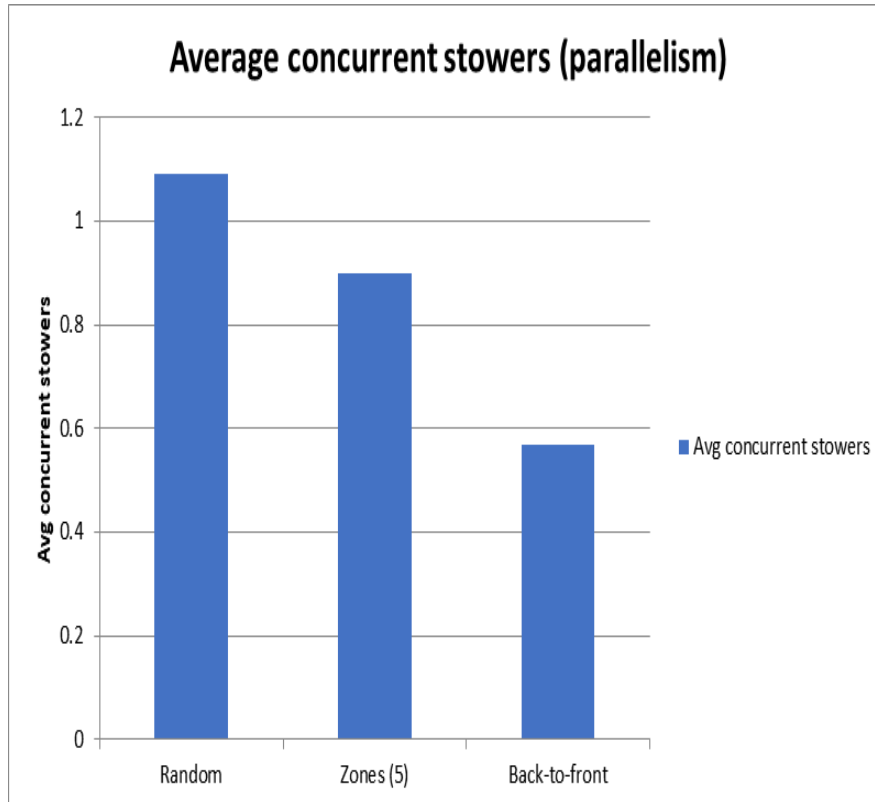


Figure 3. Average parallelism (concurrent stowing) by strategy.

Exact theory. Implementing the recurrence of Section 2.4 gives the same value as the simulation, **bit-for-bit**:

Case	T (simulation)	T (theory)	Match
Random, R=30, S=3	331	331	OK
Back-to-front, R=30, S=3	627	627	OK
Zones (5), R=30, S=3	420	420	OK
Random, R=30, S=6	466	466	OK
Back-to-front, R=30, S=6	1080	1080	OK

A brute-force sweep over $R = 1..8$, $S = 1..5$, four boarding patterns and up to 6 seats per row (25,600 cases) gave **0 mismatches**. The recurrence is therefore an exact theory of the process, not an approximation.

Clustering and the critical path. Define the *clustering* $\kappa(\pi)$ of a boarding order as the number of adjacent passenger pairs that share the same row (back-to-front maximizes κ ; random nearly minimizes it). In the recurrence, consecutive same-row passengers create a chain of $+S$ weights at a single cell ($r_j = c$), which lengthens the longest path — so T increases with clustering, consistent with all simulations. The longest-path view thus makes the mechanism explicit: **clustering stacks stowing work on one cell and lengthens the critical path.**

4. Discussion

Random boarding is fastest because it spreads passengers across rows so several rows stow simultaneously (highest parallelism, least blocking). Back-to-front is slowest because its six same-row passengers board consecutively and must stow at one cell one by one — in the theoretical model, a long chain of +S weights on each cell. The sensitivity analysis (Figure 2) confirms the gap grows with stow time S , exactly as the recurrence's +S blocking term predicts.

5. Limitations and Future Work

- Simplified model: no assigned-seat interference, no dual aisle or second door.
- The theoretical recurrence uses a naive $O(N)$ search for j ; it can be reduced to $O(N \cdot R)$ overall with preprocessing.
- Future work: prove monotonicity of T in clustering; add the "Steffen method" and window-middle-aisle boarding; test a model in which parallelism is disabled as a direct causal test.

6. Conclusion

Dispersed (random) boarding outperforms clustered (back-to-front) boarding. We explain this with a mechanism metric (parallelism) corroborated by an **exact longest-path (max-plus) model** that computes the boarding time without simulation and matches it bit-for-bit. Clustering same-row passengers stacks stowing weights on a single aisle cell and lengthens the critical path — the underlying principle behind the observed gap.

References

Steffen, J. H. (2008). Optimal boarding method for airline passengers. *Journal of Air Transport Management*. Van Landeghem, H., & Beuselinck, A. (2002). Reducing passenger boarding time in airplanes. *European Journal of Operational Research*. Bachmat, E., et al. (2006). Analysis of airplane boarding via space-time geometry and random matrix theory.